NKG DAG Retrieval Debug Report

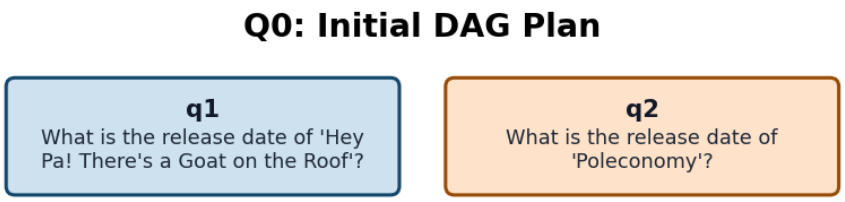
dataset: hotpotqa
engine: dag
provider: local_vllm
model: Qwen/Qwen3-30B-A3B-Instruct-2507-FP8
run-dir: /home/taluka/discourse_hyper_graph/sample_exp/nkg/hotpotqa/run_v1
questions in report: 3
retrieval args: {'top_k_seeds': 9, 'max_depth': 1, 'beam_width': 6, 'final_top_k': 20, 'mode': 'all', 'beam_mmr_lambda': 0.6, 'final_mmr_lambda': 0.6, 'search_mode': 'beam_search', 'deduplicate_paths': True, 'neutral_layout': True, 'max_fork_branches': 3}
generated: 2026-07-22T12:10:48
aggregate over this subset: EM=0.6667 F1=0.9167 n=3

Q0 [comparison / hard] 5a7555215542996c70cfaee1

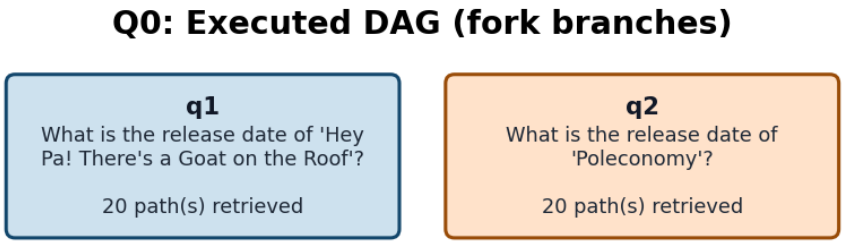
Which came to market first, "Hey Pa! There's a Goat on the Roof" or "Poleconomy"

Gold answer	Hey Pa! There's a Goat on the Roof
LLM answer	Hey Pa! There's a Goat on the Roof came to market first.
EM / F1	**0.00 / 0.75**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What is the release date of 'Hey Pa! There's a Goat on the Roof'?	20 path(s)
q2	0	—	What is the release date of 'Poleconomy'?	20 path(s)

Final context fed to generation (2000 tokens)

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Query plan:
- q1: What is the release date of 'Hey Pa! There's a Goat on the Roof'?
- q2: What is the release date of 'Poleconomy'?

Level 0:

Subquery: What is the release date of 'Hey Pa! There's a Goat on the Roof'?
"Hey Pa! There's a Goat on the Roof was issued in 1966." -> (has ) -> [hey pa! there's a goat on the roof
(Children's board game from 1966)]
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"Hey Pa! There's a Goat on the Roof was issued in 1966." -> YEAR_OF -> "Hey Pa! There's a Goat on the Roof was issued by Parker Brothers."
Logical Conclusions:
The board game 'Hey Pa! There's a Goat on the Roof' was issued by Parker Brothers in 1966.

"Hey Pa! There's a Goat on the Roof was issued in 1966." -> YEAR_OF -> "Hey Pa! There's a Goat on the Roof was a children's board game."
Logical Conclusions:
The children's board game 'Hey Pa! There's a Goat on the Roof' was issued in 1966.

[hey pa! there's a goat on the roof (Children's board game from 1966)] -> (participates in event: Hey Pa! There's a Goat on the Roof was issued by Parker Brothers.) -> "Hey Pa! There's a Goat on the Roof was issued by Parker Brothers."

[hey pa! there's a goat on the roof (Children's board game from 1966)] -> (participates in event: Hey Pa! There's a Goat on the Roof was a children's board game.) -> "Hey Pa! There's a Goat on the Roof was a children's board game."

"He Will Have His Way was released on 12 November 2010." -> (has) -> [Finn Brothers Compilation Albums (Compilation albums of Finn Brothers songs)]

"He Will Have His Way was released on 12 November 2010." -> RELEASE_DATE_OF -> "He Will Have His Way is a compilation album featuring male Australian and New Zealand musicians performing songs written by Neil Finn and Tim Finn (The Finn Brothers)."
Logical Conclusions:
'He Will Have His Way' was released on 12 November 2010.

"He Will Have His Way was released on 12 November 2010." -> FOLLOWS -> "He Will Have His Way is a sequel to 'She Will Have Her Way'."
Logical Conclusions:
'He Will Have His Way' was released in 2010, while its predecessor 'She Will Have Her Way' was released in 2005.

"He Will Have His Way was released on 12 November 2010." -> PART_OF -> "The two albums were later released together with extra tracks as the double CD 'They Will Have Their Way'."
Logical Conclusions:
'He Will Have His Way' was later released together with 'She Will Have Her Way' as the double CD 'They Will Have Their Way'.

"He Will Have His Way was released on 12 November 2010." -> COMPANION_TO -> "'She Will Have Her Way' is a 2005 album featuring female Australian and New Zealand musicians."
Logical Conclusions:
'She Will Have Her Way' and 'He Will Have His Way' are companion albums featuring female and male artists respectively covering songs by the Finn Brothers.

"Both 'DIVEBAR Days Revisited' and 'A Sinners Christmas 2' were released in 2013." -> (has) -> [Zachary Throne Related Works (TV series and songs associated with Zachary Throne)]

"Both 'DIVEBAR Days Revisited' and 'A Sinners Christmas 2' were released in 2013." -> SUPPORTS -> "'Looking Through Your Eyes' was featured on LeAnn Rimes' album 'Sittin' on Top of the World'."
Logical Conclusions:
'Looking Through Your Eyes' was featured on LeAnn Rimes' album 'Sittin' on Top of the World'.

"Both 'DIVEBAR Days Revisited' and 'A Sinners Christmas 2' were released in 2013." -> YEAR_OF -> "With the Sin City Sinners, Zachary Throne recorded the album 'DIVEBAR Days Revisited'."
Logical Conclusions:
'DIVEBAR Days Revisited' was released in 2013.

"Both 'DIVEBAR Days Revisited' and 'A Sinners Christmas 2' were released in 2013." -> (has) -> [a sinners christmas 2 (Album recorded with Sin City Sinners)]

"Both 'DIVEBAR Days Revisited' and 'A Sinners Christmas 2' were released in 2013." -> YEAR_OF -> "With the

Sin City Sinners, Zachary Throne recorded the album 'A Sinners Christmas 2'."
Logical Conclusions:
'A Sinners Christmas 2' was released in 2013.

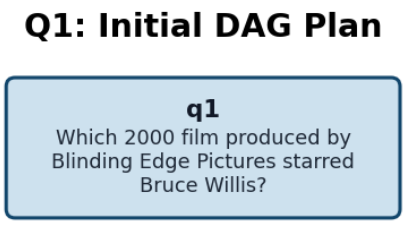
[pashmina goat (breed providing pashmina wool)] -> (participates in event: The Chyangara go …[truncated for report]

Q1 [bridge / hard] 5a8b3af455429971feec467b

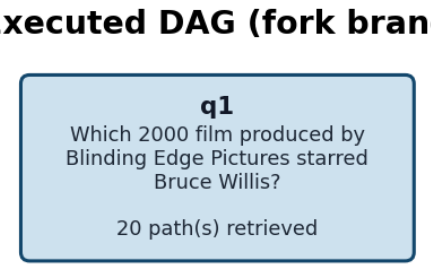
Blinding Edge Pictures produced which 2000 film starring Bruce Willis?

Gold answer	Unbreakable
LLM answer	Unbreakable
EM / F1	**1.00 / 1.00**
Fallback mode	single-node plan — delegated straight to the wrapped engine (no DAG structure)

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Which 2000 film produced by Blinding Edge Pictures starred Bruce Willis?	20 path(s)

Final context fed to generation (1006 tokens)

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"Blinding Edge Pictures produced 'Unbreakable' (2000)." -> (has ) -> [M. Night Shyamalan and Related Films (Films including Unbreakable and Lay the Favorite)] -> (participates in event: 'Unbreakable' stars Bruce Willis.) -> "'Unbreakable' stars Bruce Willis." -> FEATURES -> "'Unbreakable' is a 2000 American superhero thriller film."
Logical Conclusions:
Bruce Willis stars in the film 'Unbreakable'.

[blinding edge pictures (Production company for Signs)] -> (participates in event: Blinding Edge Pictures produced 'Unbreakable' (2000).) -> "Blinding Edge Pictures produced 'Unbreakable' (2000)."

"'Unbreakable' stars Bruce Willis." -> CO_STARS -> "'Unbreakable' stars Robin Wright."
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Logical Conclusions:
Both facts list stars of the film 'Unbreakable'.

"'Unbreakable' stars Bruce Willis." -> (has) -> [bruce willis (Actor starring in multiple films)] -> (participates in event: Mercury Rising stars Bruce Willis and Alec Baldwin.) -> "Mercury Rising stars Bruce Willis and Alec Baldwin."

[bruce willis (Actor starring in multiple films)] -> (participates in event: The two veteran NYPD partners are played by Willis and Morgan.) -> "The two veteran NYPD partners are played by Willis and Morgan."

[bruce willis (Actor starring in multiple films)] -> (participates in event: In Country stars Bruce Willis and Emily Lloyd.) -> "In Country stars Bruce Willis and Emily Lloyd."

[bruce willis (Actor starring in multiple films)] -> (participates in event: Striking Distance stars Bruce Willis.) -> "Striking Distance stars Bruce Willis."

[bruce willis (Actor starring in multiple films)] -> (participates in event: Bruce Willis earned a best supporting actor Golden Globe nomination for his role) -> "Bruce Willis earned a best supporting actor Golden Globe nomination for his role in In Country."

"'Unbreakable' stars Bruce Willis." -> PLOT_SUMMARY -> "In 'Unbreakable', Dunn explores and reluctantly confronts his powers while trying to navigate a difficult family life."
Logical Conclusions:
In Unbreakable, Dunn explores and reluctantly confronts his powers while trying to navigate a difficult family life.

[bruce willis (Actor starring in multiple films)] -> (participates in event: Lay the Favorite stars Bruce Willis, Rebecca Hall, Catherine Zeta-Jones and Josh) -> "Lay the Favorite stars Bruce Willis, Rebecca Hall, Catherine Zeta-Jones and Joshua Jackson."

"'Unbreakable' stars Bruce Willis." -> APPEARS_IN -> "In 'Unbreakable', a security guard named David Dunn survives a horrific train crash."
Logical Conclusions:
Fact 1 identifies a star of 'Unbreakable', while fact 10 describes the plot involving that star's character.

"'Unbreakable' stars Bruce Willis." -> PLOT_SUMMARY -> "After the train crash in 'Unbreakable', with the help of a manipulative disabled comic book shop owner named Elijah Price, David Dunn learns that he possesses superhuman powers."
Logical Conclusions:
After the train crash in Unbreakable, with the help of a manipulative disabled comic book shop owner named Elijah Price, David Dunn learns that he possesses superhuman powers.

[M. Night Shyamalan and Related Films (Films including Unbreakable and Lay the Favorite)] -> (participates in event: Lay the Favorite stars Bruce Willis, Rebecca Hall, Catherine Zeta-Jones and Josh) -> "Lay the Favorite stars Bruce Willis, Rebecca Hall, Catherine Zeta-Jones and Joshua Jackson." -> SAME_ACTOR -> "In Mercury Rising, Bruce Willis plays Art Jeffries."
Logical Conclusions:
Both facts involve Bruce Willis starring in different films.

"Lay the Favorite stars Bruce Willis, Rebecca Hall, Catherine Zeta-Jones and Joshua Jackson." -> SAME_ACTOR -> "In Country stars Bruce Willis and Emily Lloyd."
Logical Conclusions:
Both facts involve Bruce Willis starring in different films.

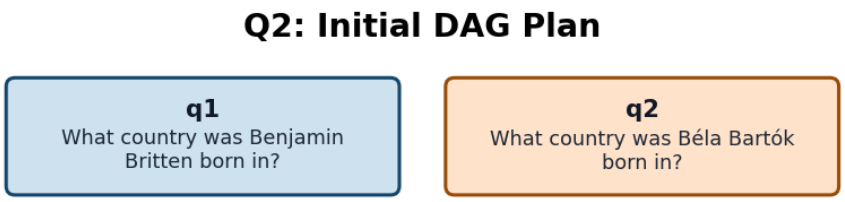
"Lay the Favorite stars Bruce Willis, Rebecca Hall, Catherine Zeta-Jones and Joshua Jackson." -> SAME_ACTOR -> "Mercury Rising stars Bruce Willis and Alec Baldwin."
Logical Conclusions:
Both facts involve Bruce Willis starring …[truncated for report]

Q2 [comparison / hard] 5ae12bcc5542997b2ef7d11a

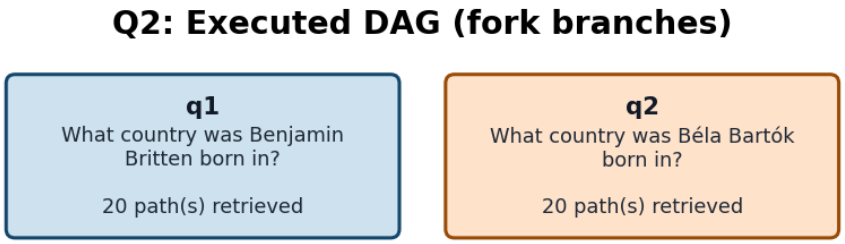
Are Benjamin Britten and Béla Bartók from the same country?

Gold answer	no
LLM answer	No
EM / F1	**1.00 / 1.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What country was Benjamin Britten born in?	20 path(s)
q2	0	—	What country was Béla Bartók born in?	20 path(s)

Final context fed to generation (2000 tokens)

```
Query plan:
- q1: What country was Benjamin Britten born in?
- q2: What country was Béla Bartók born in?

Level 0:

Subquery: What country was Benjamin Britten born in?
"Benjamin Britten was born on 22 November 1913." -> BIRTH_DATE -> "Edward Benjamin Britten, Baron Britten
was an English composer, conductor and pianist."
Logical Conclusions:
Benjamin Britten was born on 22 November 1913.
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"Benjamin Britten was born on 22 November 1913." -> (has) -> [Benjamin Britten (English composer and conductor)]

"Benjamin Britten was born on 22 November 1913." -> PRECEDES -> "Benjamin Britten died on 4 December 1976."
Logical Conclusions:
Britten's birth date precedes his death date, establishing his lifespan.

[Benjamin Britten (English composer and conductor)] -> (participates in event: Edward Benjamin Britten, Baron Britten was an English composer, conductor and pi) -> "Edward Benjamin Britten, Baron Britten was an English composer, conductor and pianist."

[Benjamin Britten (English composer and conductor)] -> (participates in event: Benjamin Britten was a central figure of 20th-century British classical music.) -> "Benjamin Britten was a central figure of 20th-century British classical music."

[Benjamin Britten (English composer and conductor)] -> (participates in event: The English composers whose works were premiered and recorded by the Zorian Quar) -> "The English composers whose works were premiered and recorded by the Zorian Quartet include Benjamin Britten and Michael Tippett."

[Benjamin Britten (English composer and conductor)] -> (participates in event: Benjamin Britten's best-known works include the orchestral piece.) -> "Benjamin Britten's best-known works include the orchestral piece."

[Benjamin Britten (English composer and conductor)] -> (participates in event: Notable composers who have been influenced by Bartók's string quartets include B) -> "Notable composers who have been influenced by Bartók's string quartets include Benjamin Britten."

"He lived in Berlin-Lankwitz, Germany." -> (has) -> [Germany (Country in Europe)]

"He lived in Berlin-Lankwitz, Germany." -> (has) -> [Various cities (Cities in Europe)]

"He lived in Berlin-Lankwitz, Germany." -> (has) -> [He (person living in Berlin-Lankwitz)]

"He lived in Berlin-Lankwitz, Germany." -> SUPPORTS -> "Paul Matte lived in Berlin-Lankwitz, Germany."
Logical Conclusions:
Paul Matte lived in Berlin-Lankwitz, Germany, and he lived in Berlin-Lankwitz, Germany.

"He lived in Berlin-Lankwitz, Germany." -> RESIDED_AT -> "Karl Hermann Wichelhaus was a German chemist."
Logical Conclusions:
Karl Hermann Wichelhaus lived in Berlin-Lankwitz, Germany, which is the same location mentioned for another individual.

[Nick Robinson (Actor in Frenemies)] -> (participates in event: In 1859, Orville E. Thompson helped lead the large-scale manufacturing of leathe) -> "In 1859, Orville E. Thompson helped lead the large-scale manufacturing of leather boots and shoes in Chicago."

[Nick Robinson (Actor in Frenemies)] -> (participates in event: Frenemies features an ensemble cast starring Nick Robinson.) -> "Frenemies features an ensemble cast starring Nick Robinson."

[brisbane (host city of the games)] -> (participates in event: The 2001 Goodwill Games was held in Brisbane, Queensland, Australia.) -> "The 2001 Goodwill Games was held in Brisbane, Queensland, Australia."

[Paul Tibbitt (Showrunner of SpongeBob SquarePants)] -> (participates in event: Writer Paul Tibbitt acted as the supervising/co-executive producer and showrunne) -> "Writer Paul Tibbitt acted as the supervising/co-executive producer and showrunner for the fourth season of 'SpongeBob SquarePants'."

[John Pattison (Editorial staff at Barrie & Rockliff)] -> (participates in event: Barrie & Rockliff's editorial staff included John Pattison.) -> "Barrie & Rockliff's editorial staff included John Pattison."

[Sebastian Shaw (Reader in The Great War documentary)] -> (participates in event: Readings in 'The Great War' were …[truncated for report]